ECM3412/ECMM409
Nature-Inspired Computation

Introduction to the Module

Module Convenors

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(Please email either of us and include “ECM3412/ECMM409” in the subject)

Module Information

- Lectures
  - Lecture 1: Monday 14:35, Harrison 004
  - Lecture 2: Thursday 11:35, Harrison 004
- Workshops
  - Week 3 & Week 10
  - Monday/Tuesday/Thursday (check timetable for your groups)
  - Assessment

CA	Type of Task	Submission Deadline	Worth
CA1	Programming Task & Report	11 Nov 2025 for ECM3412 23 Oct 2025 for ECMM409	40%
CA2	Team Project (ECMM409 Only)	26 Nov 2025	60%
Exam	(ECM3412 Only)	May 2026	60%

- Study resources include lecture slides, module descriptors, suggested reading, CA, etc… All on ELE

What is Nature-Inspired Computation?

- To understand this, we can split the title into its component parts:
  - "Nature" or what elements of nature are considered in NIC?
  - "Inspired" or how can nature influence computing?
  - "Computation" or what problems can be solved by NIC?

“Nature”

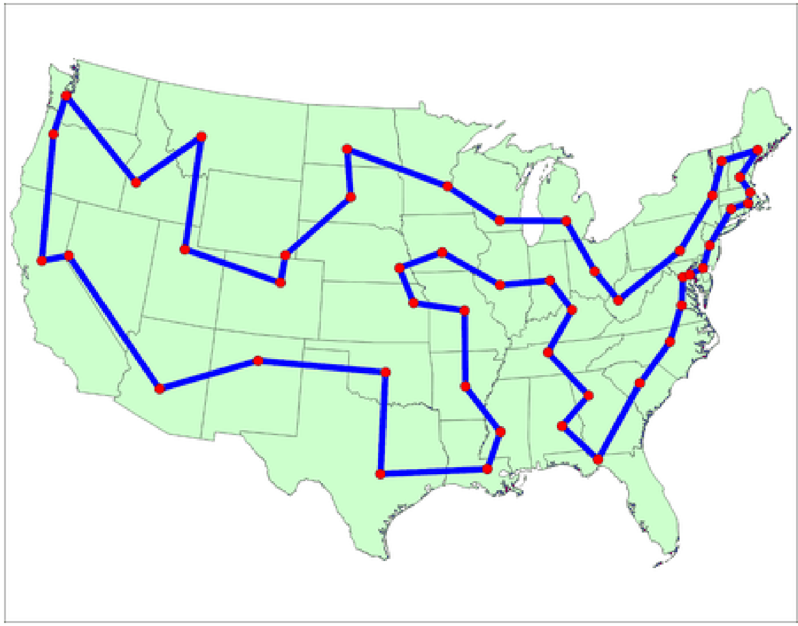
- Which natural systems can benefit computing?
  - Evolution
  - Brains
  - Collective behaviours (e.g. ant colonies, swarms)

“Inspired”

- Why should natural systems be used as inspiration for new algorithms?
  - Evolution – created some of the most complex objects we know of – ourselves.
  - Brains – we solve many seemingly difficult problems
  - Collective behaviours – a collection of seemingly unintelligent individuals can display intelligence as part of a collective

“Computation”

- What problems can be solved by using nature inspired computation?
  - Some tasks that organisms have had to evolve to be good at, are also tasks we’re interested in in computing
    - Recognising patterns
    - Finding shortest paths
    - Searching (e.g. for food etc..)



Advantages of Nature-Inspired Computing

- Nature-Inspired techniques tend to produce good results in reasonable computational time on a vast array of real-world problems

Examples:

- Evolutionary algorithms are able to optimise complex systems on relatively modest hardware – generally better than classical optimisation methods.
- Artificial neural networks are generally more successful than the classical methods for pattern recognition.
- Swarm intelligence methods are good at modelling complex systems emerging from collective behaviour of simple agents.

Natural Systems

- Throughout the module we will see:
  - **Ants** that can help us timetable lectures
  - **Swarms of particles** that solve hugely complex problems
  - **Artificial neural networks** that can learn patterns in data for themselves

- All created by one biological process – **evolution!**

Syllabus

Weeks 1—5 Evolutionary Algorithms & Ant Colony Optimisation

1-intro, 2-complexity, 3-problems & fitness landscapes, 4-genetic operators, 5-encodings/applications, 6-genetic programming (2), 8-ant colony optimisation (3)

Workshop: Genetic Algorithms

Week 6: Reading Week

Weeks 7 - 9 Swarm Intelligence & Multi-Objective Methods

11-flocking behaviours, 12-multi-agent systems, 13-particle swarm optimisation, 14-multiobjective genetic algorithms (2) , 16-multiobjective swarm intelligence

Weeks 10 and 11 Neural Networks & Artificial Life

17-neural networks, 18-neuromorphic computing, 19-self-organising maps, 20-artificial life & cellular automata

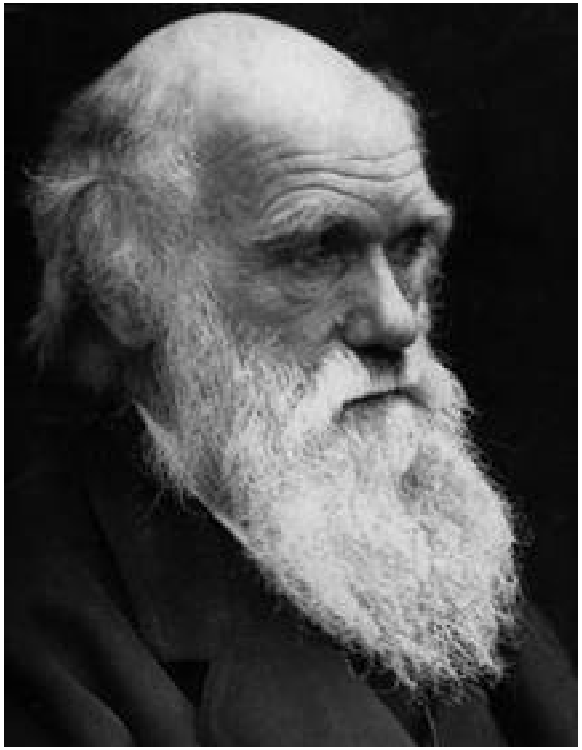
Workshop: Neural Networks

Revision Lecture & Model Exam (Closer to Exams)

Natural Evolution

- *"As many more individuals of each species are born than can possibly survive; and as, consequently, there is a frequently recurring struggle for existence, it follows that any being, if it vary however slightly in any manner profitable to itself, under the complex and sometimes varying conditions of life, will have a better chance of surviving, and thus be naturally selected. From the strong principle of inheritance, any selected variety will tend to propagate its new and modified form."*

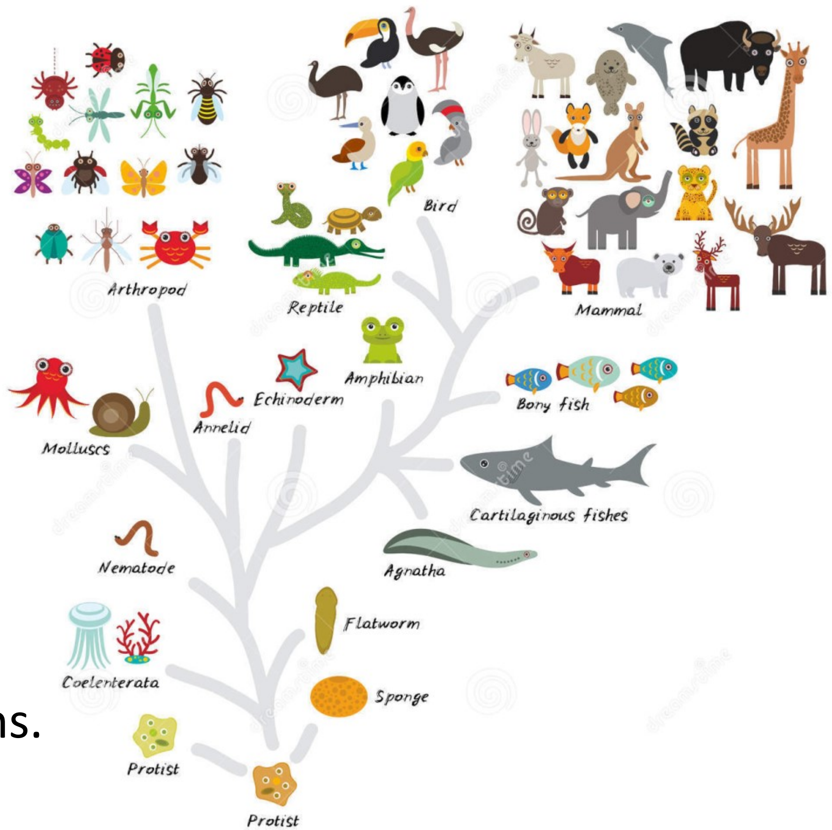
Charles Darwin, The Origin of Species (1859)



Evolution/Survival of the Fittest

- In particular, any new mutation that appears in a child (e.g. longer neck, longer legs, thicker skin, longer gestation period, bigger brain, etc etc) and which *helps* it in its efforts to survive long enough to have children, will become more and more widespread in future generations.

- The theory of evolution is the statement that all species on Earth have arisen in this way by evolution from one or more very simple self-reproducing molecules in the primeval soup (tree of life).



- We have evolved via the accumulation of countless advantageous (in context) mutations over countless generations, and species have diversified to occupy niches, as a result of different environments favouring different mutations.

Natural Evolution as a Problem Solving Method

All the complexity in the natural world has evolved from a single common ancestor. How?

The theory is: given:

1. A population of organisms which can reproduce in a challenging/changing environment
2. A way of continually generating diversity in new `child' organisms

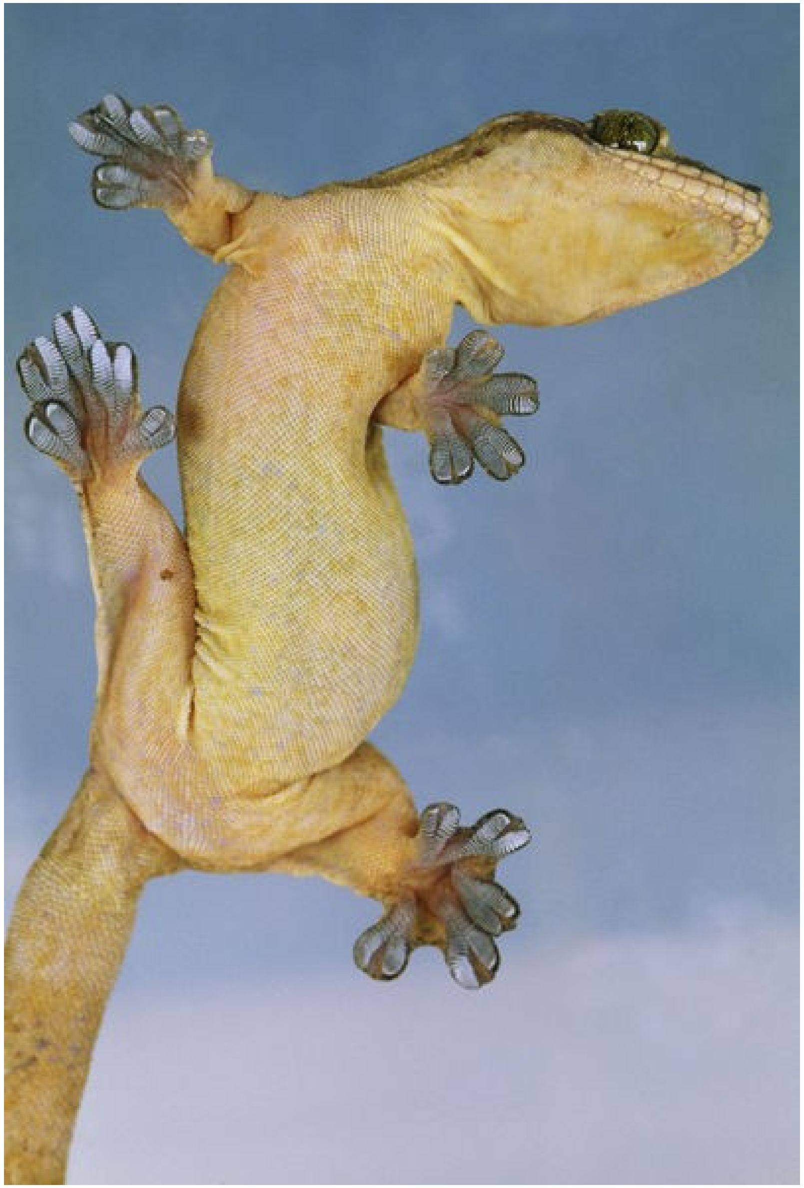
A `survival of the fittest principle will naturally emerge: organisms which tend to have healthy, fertile children will dominate.

Evolution as Problem Solving

Here's a problem:

Design a material for the soles of boots that can help you walk up a smooth vertical brick wall

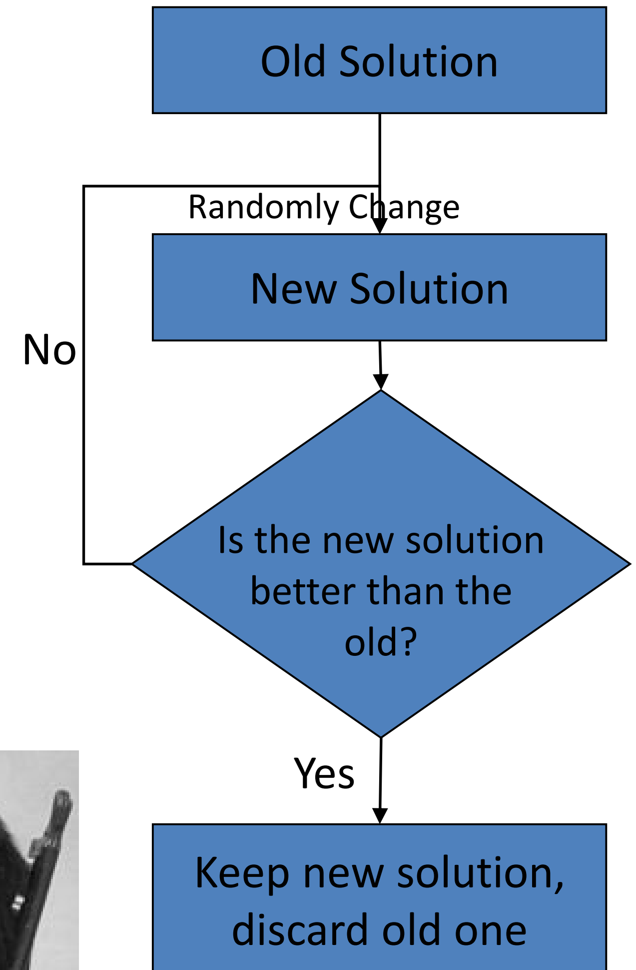
We haven't solved this, but nature has: Geckos



Evolution as a Problem Solving Method

Can view evolution as a way of solving the problem: *How can I survive in this environment?*

The basic *method* of it is trial and error.
i.e. evolution is in the family of methods that do something like this →



But this appears to be a recipe for problem solving algorithms which take forever, with little or no eventual success.

Infinite Monkey Theorem:
Given an infinite length of time, a chimpanzee punching at random on a typewriter would almost surely type out all of Shakespeare's plays.

The Magic Ingredients

Although randomness (stochasticity) is involved in an evolutionary algorithm, other ingredients, inspired by nature are required to develop a functioning algorithm.

Ingredient 1: Use a population of organisms that are competing for resources.

Ingredient 2: Select `parents' with a relatively *weak bias* towards the fittest.

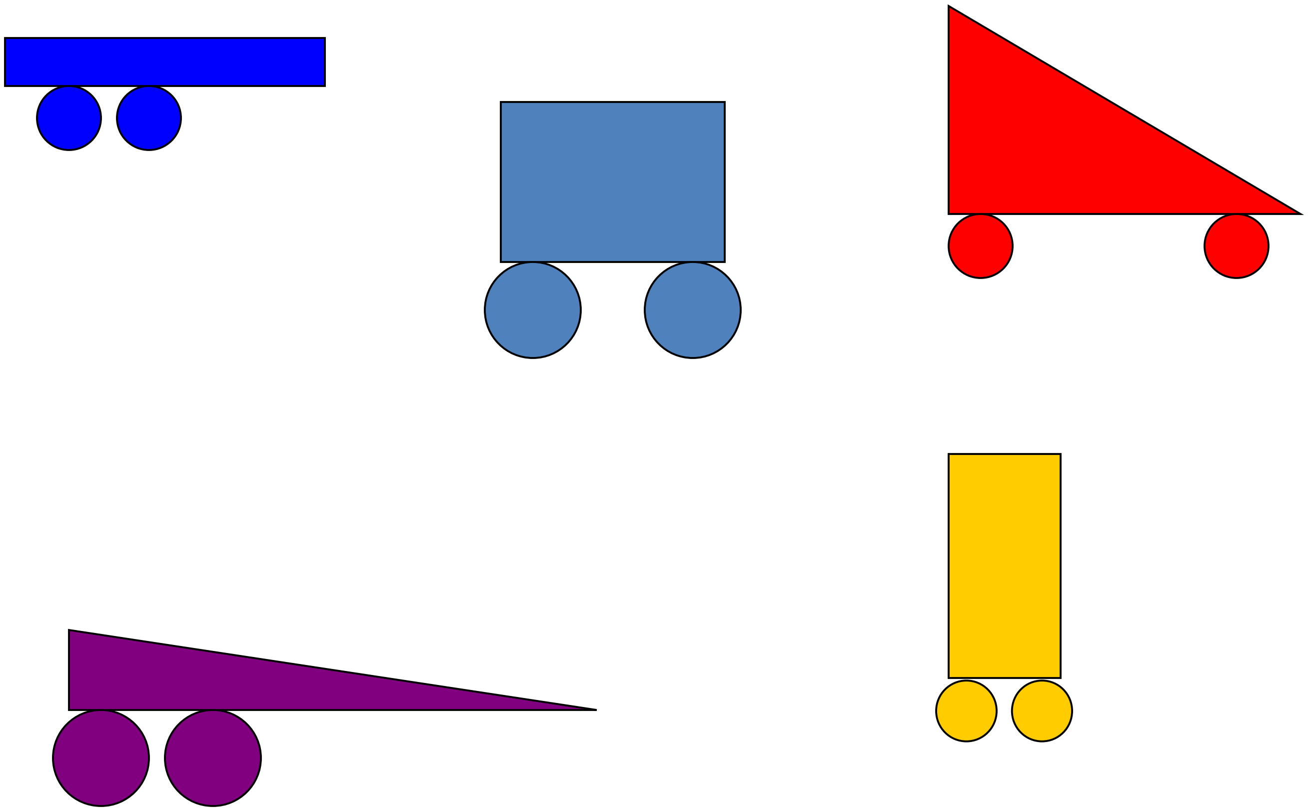
It's not really plain survival of the fittest, what works is
the fitter you are, the more chance you have to reproduce,
and it works best if even the least fit still have some chance.

Mutate these, i.e. apply a small change.

Ingredient 3: It can sometimes help to use *recombination* of two or more `parents' – I.e. generate new candidate solutions by combining bits and pieces from different previous solutions.

1&2 are required. 3 is optional, but is often helpful...

Initial population



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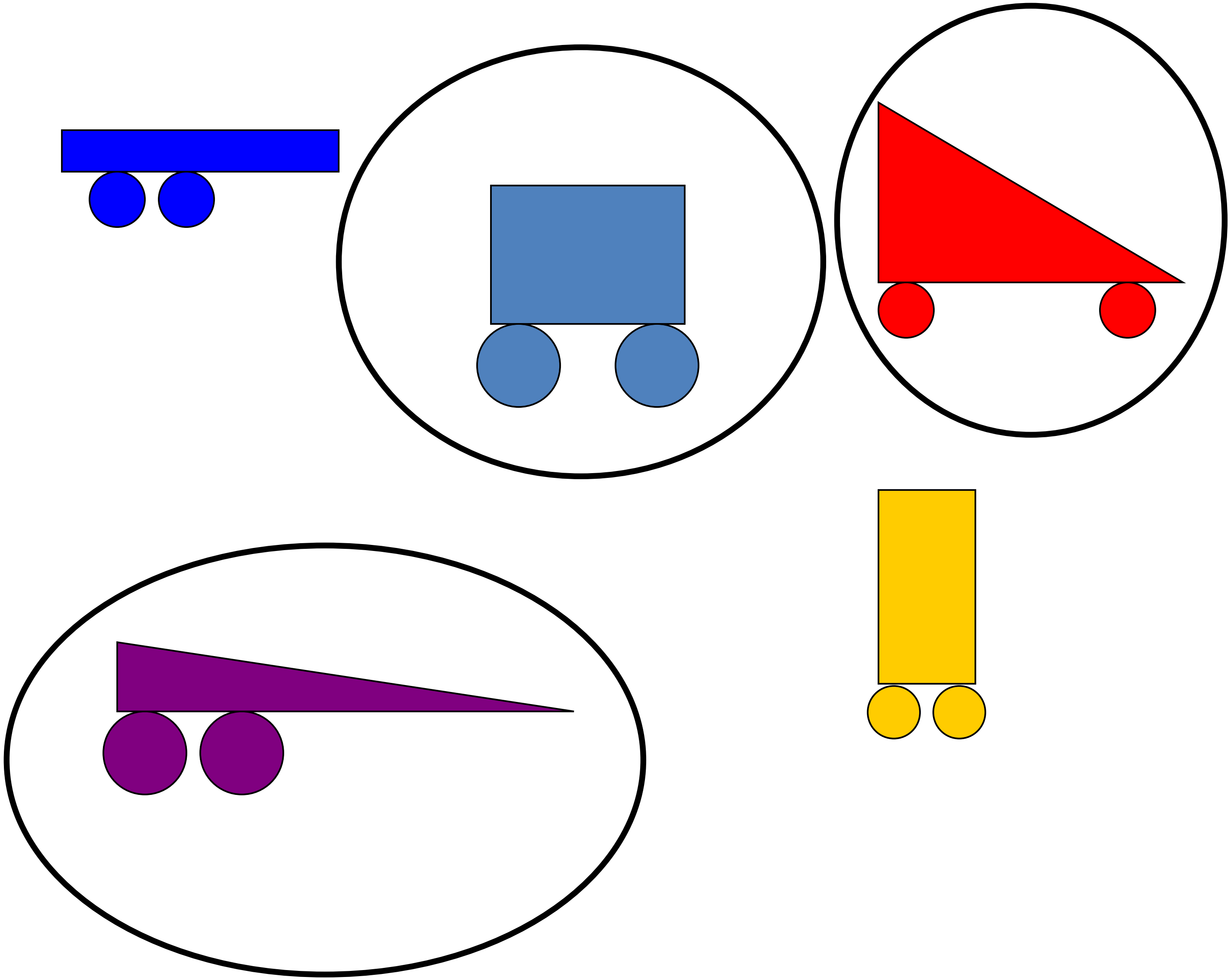
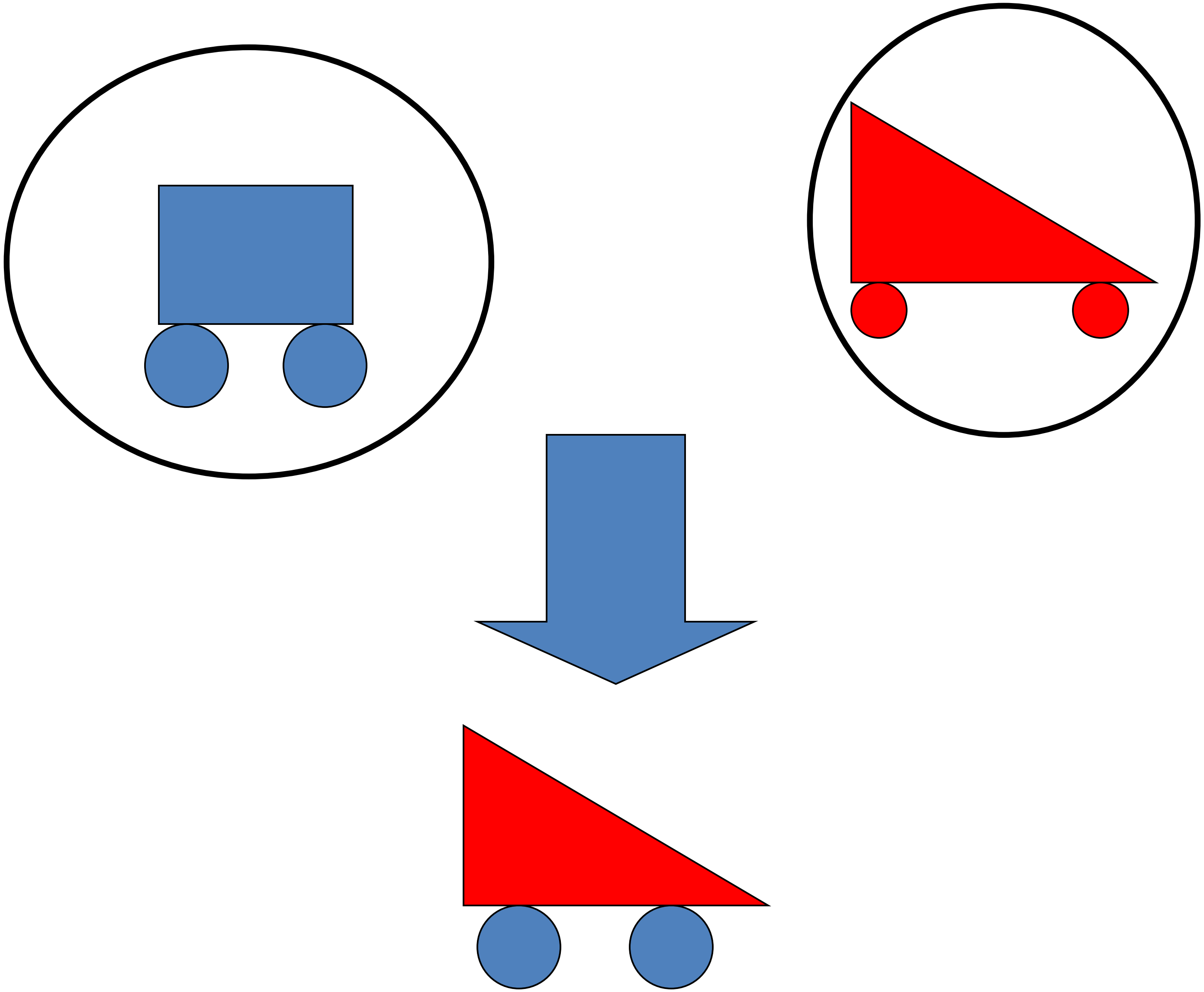
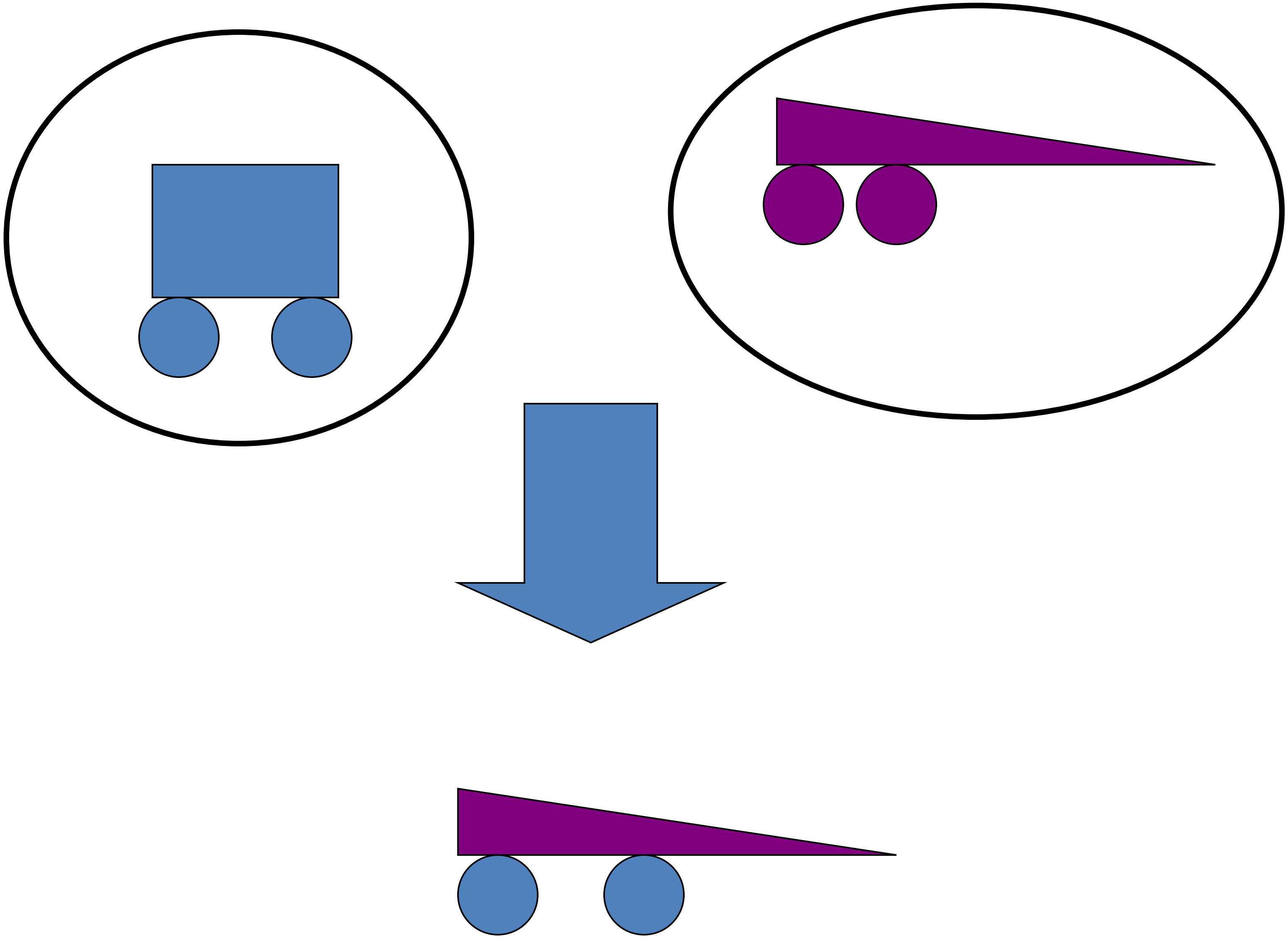


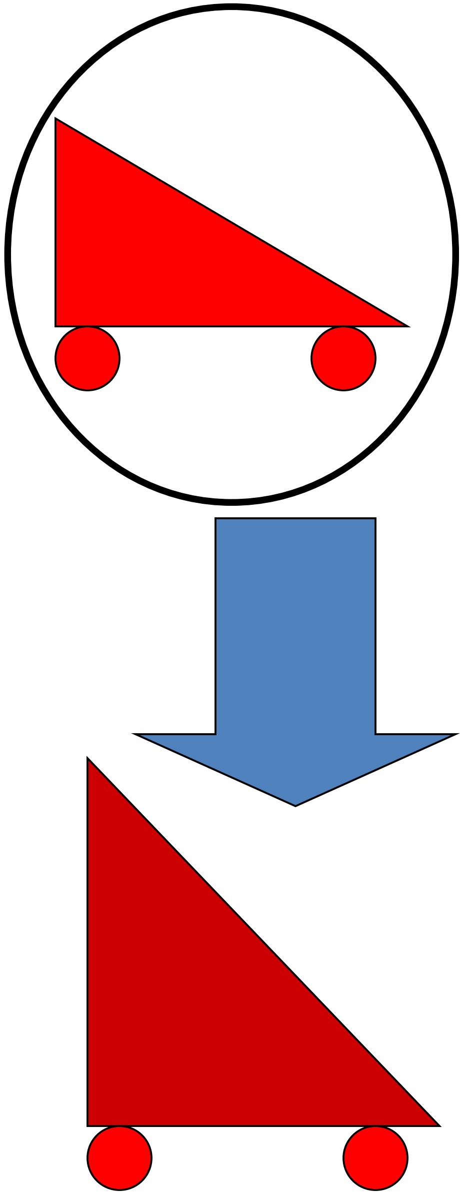
Crossover



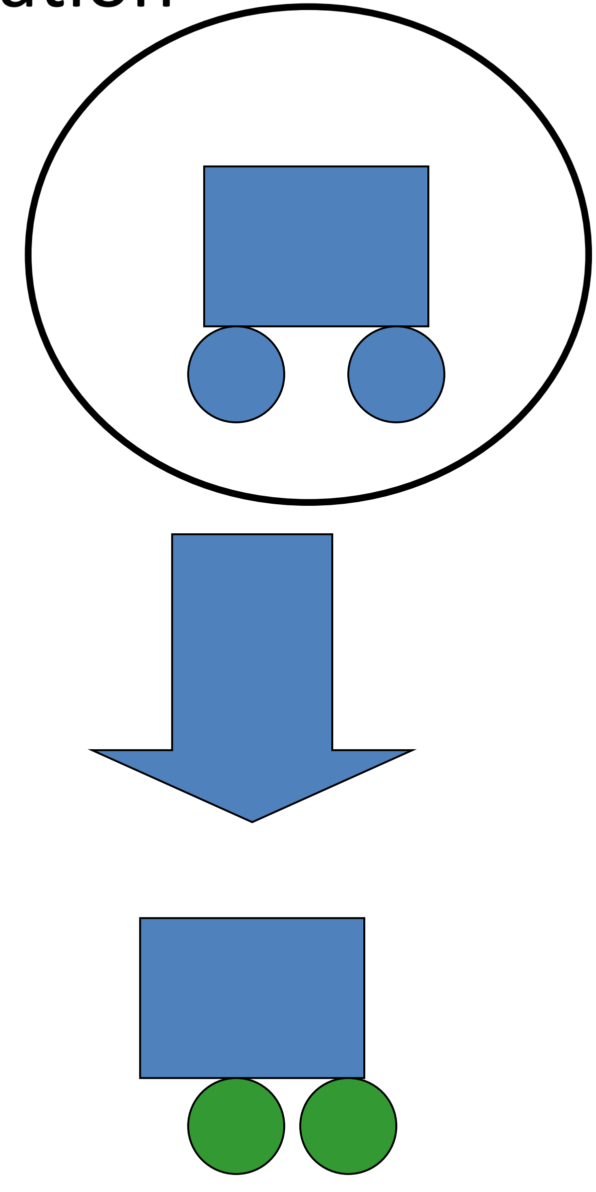
Another Crossover



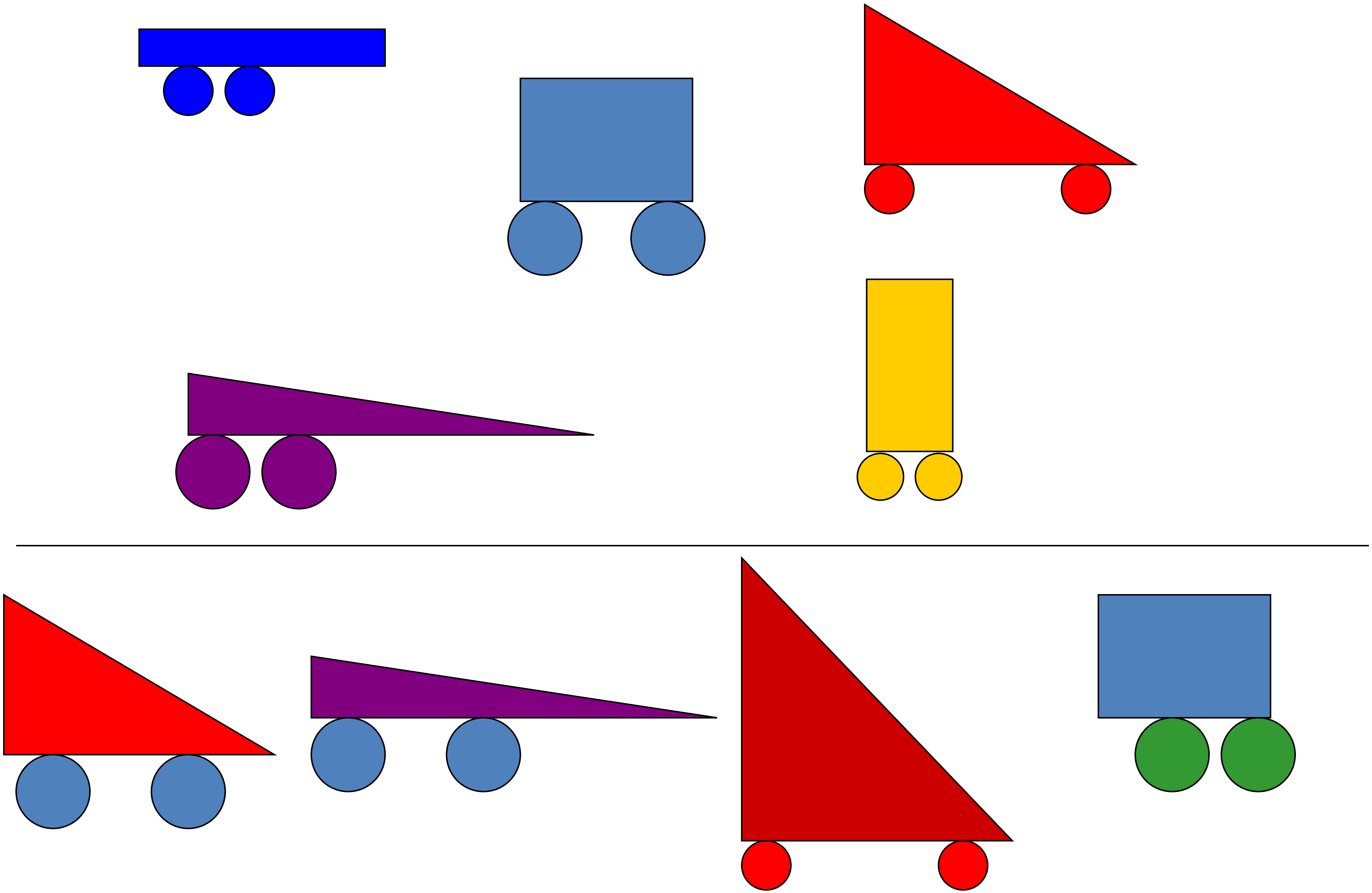
A mutation



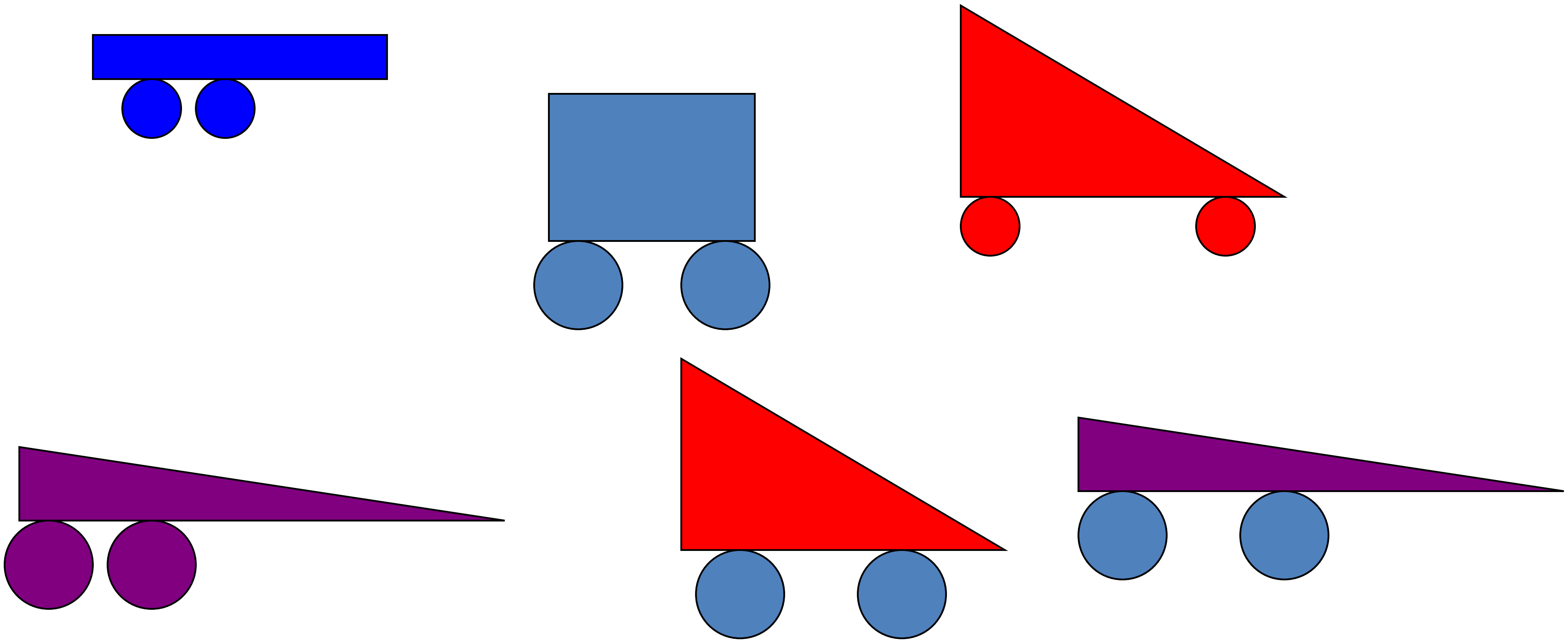
Another Mutation



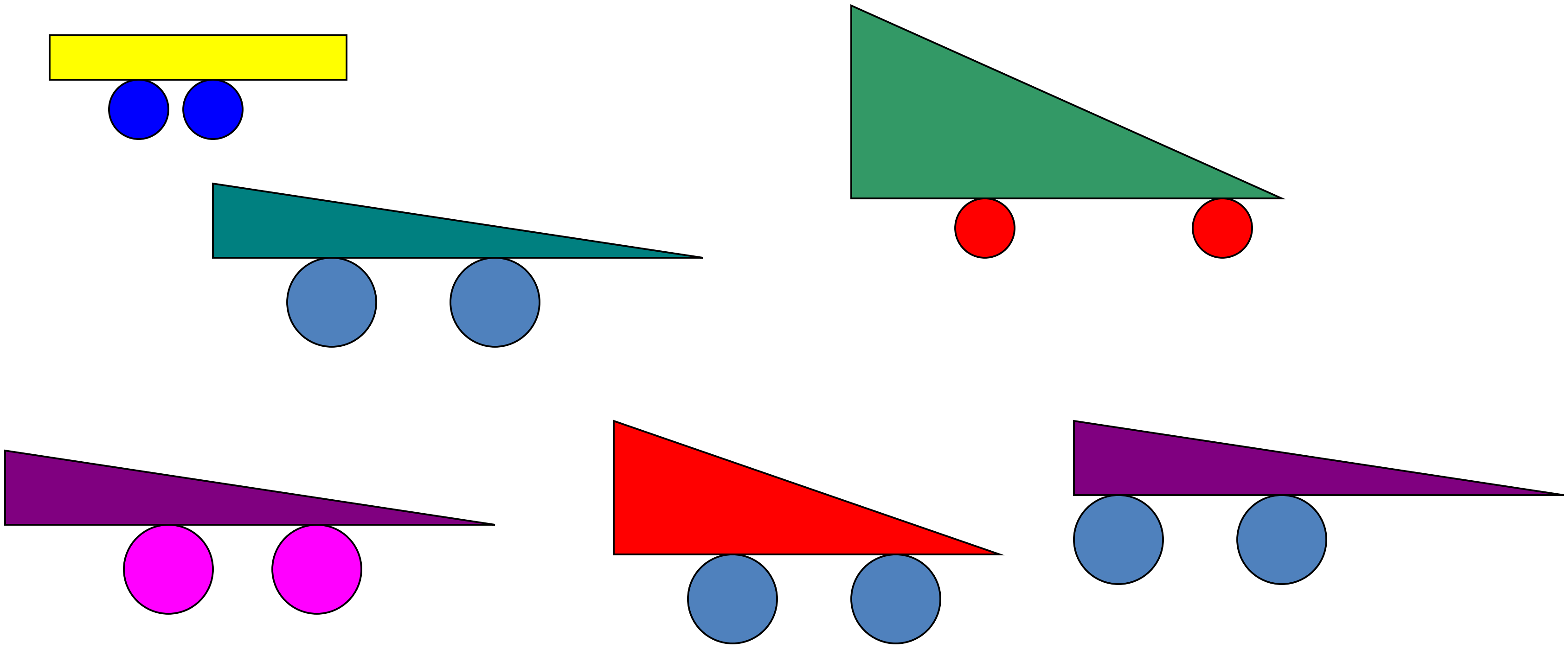
Old population + children



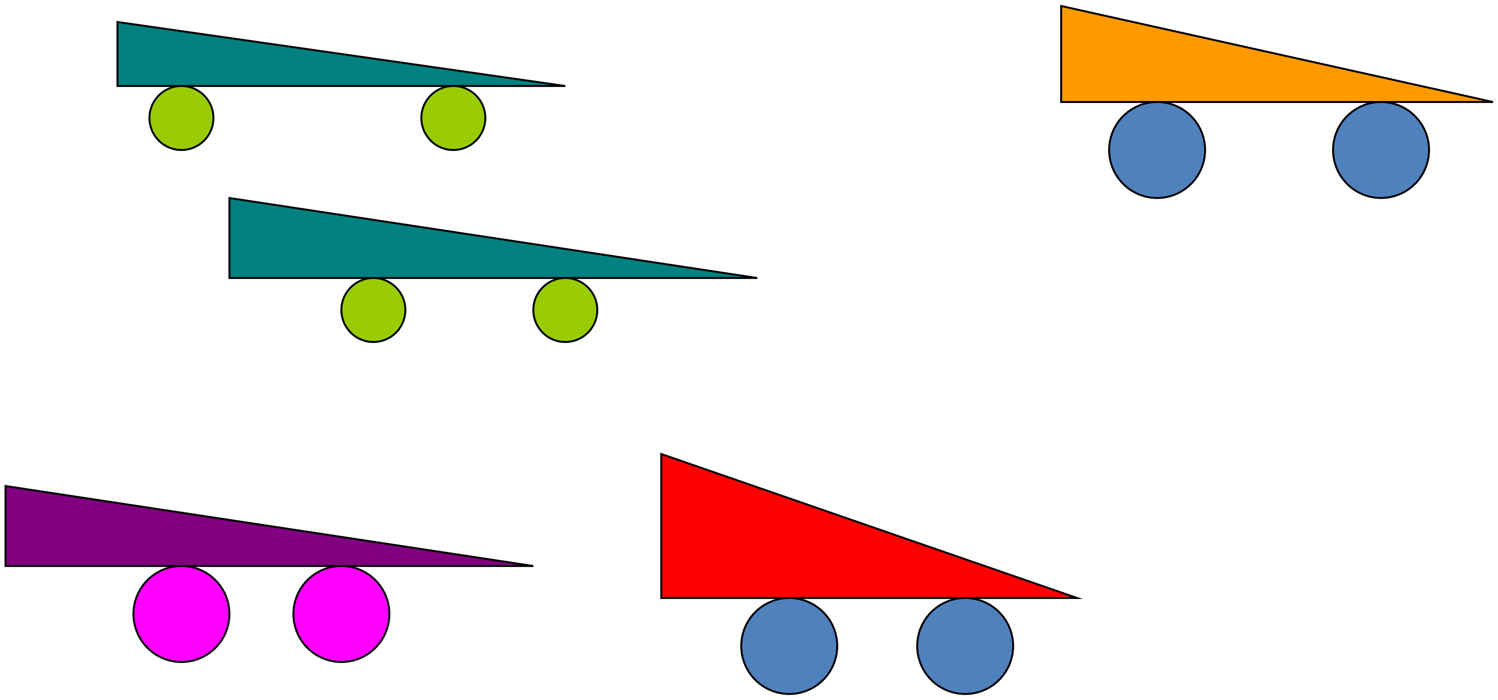
New Population: Generation 2



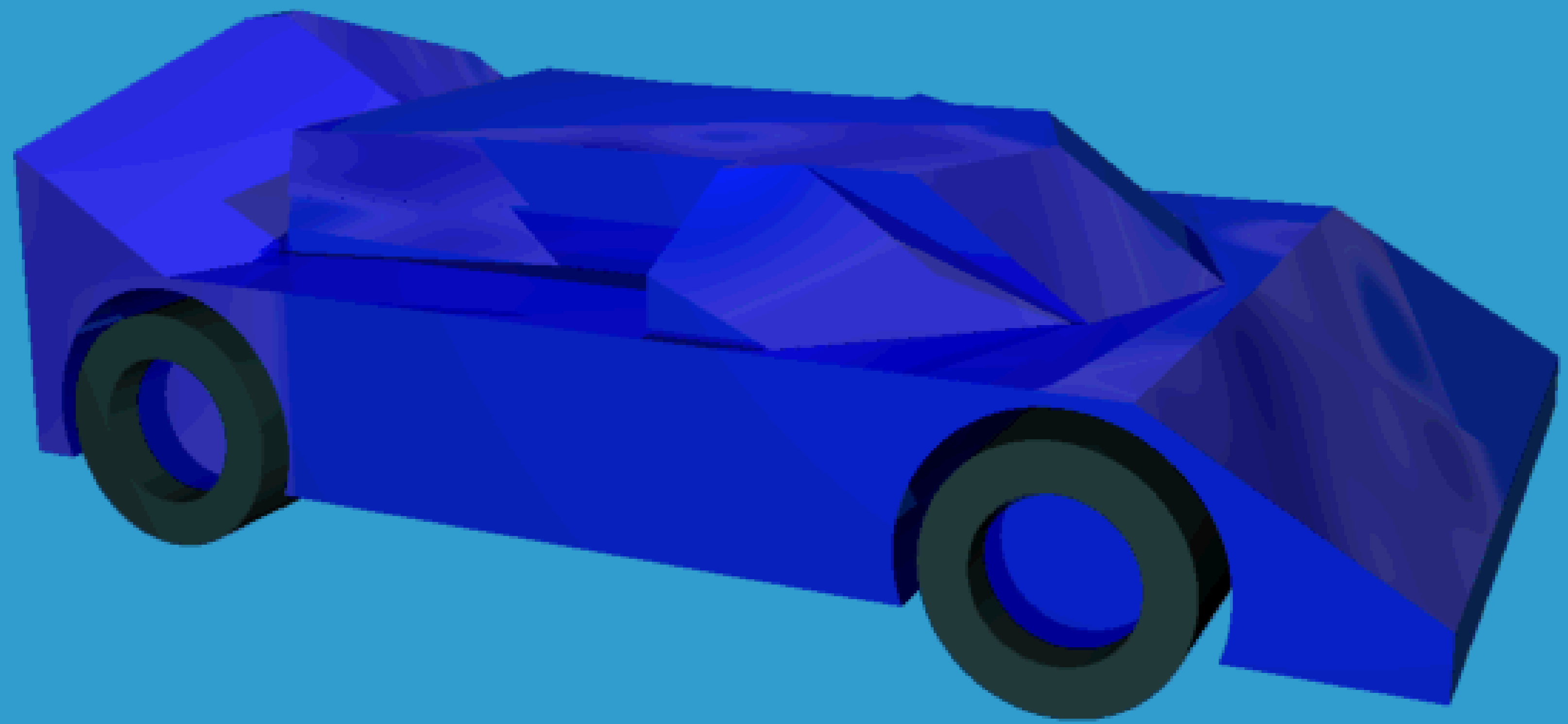
Generation 3



Generation 4, etc …

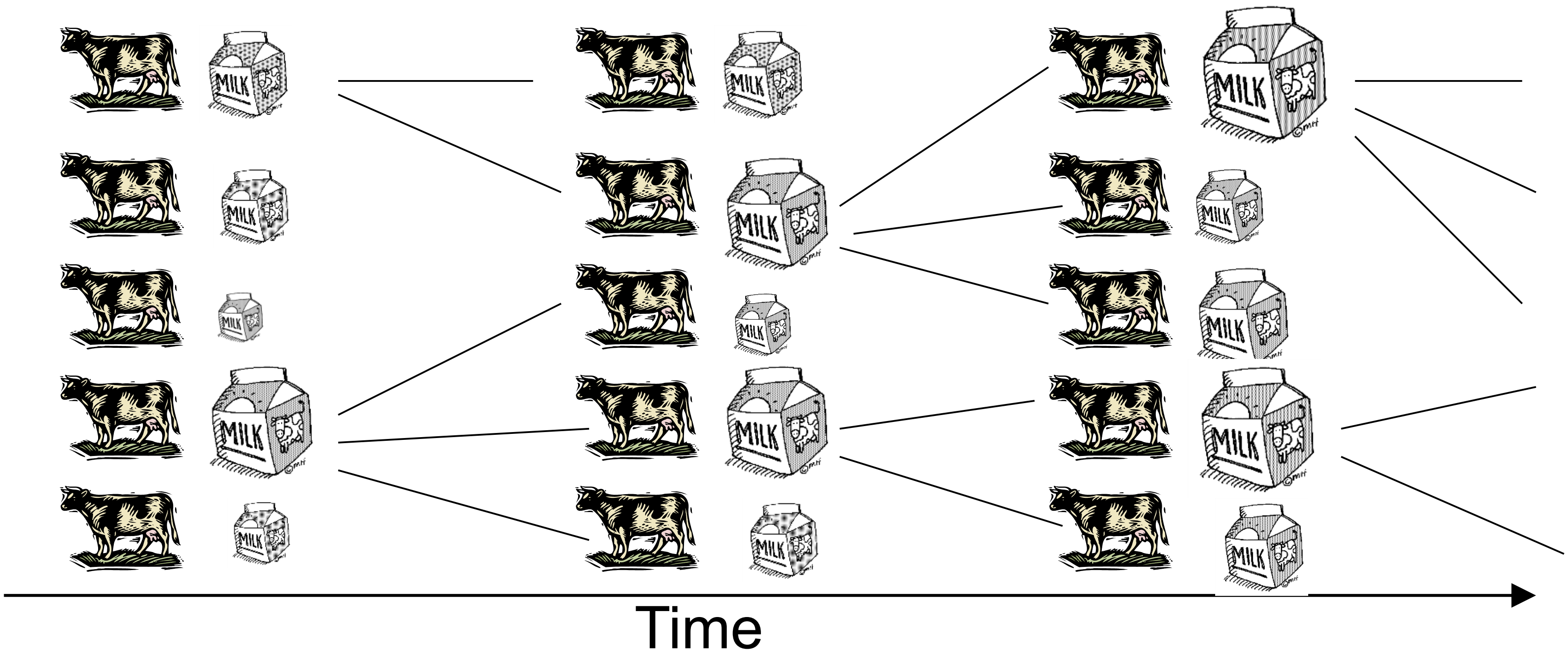


Bentley’s thesis work

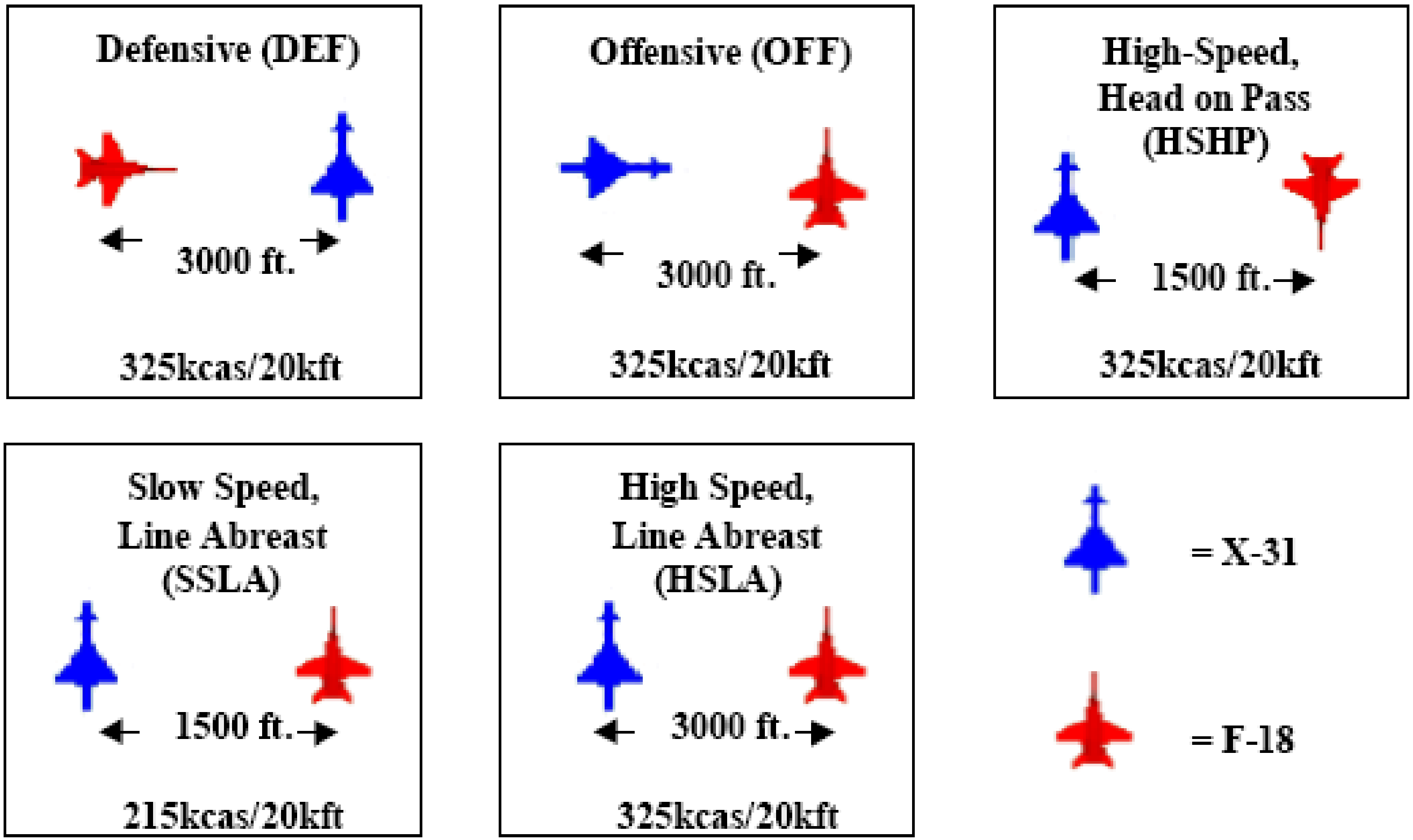


Fixed wheel positions, constrained bounding area,
Chromosome is a series of slices
Fitnesses evaluated via a simple airflow simulation

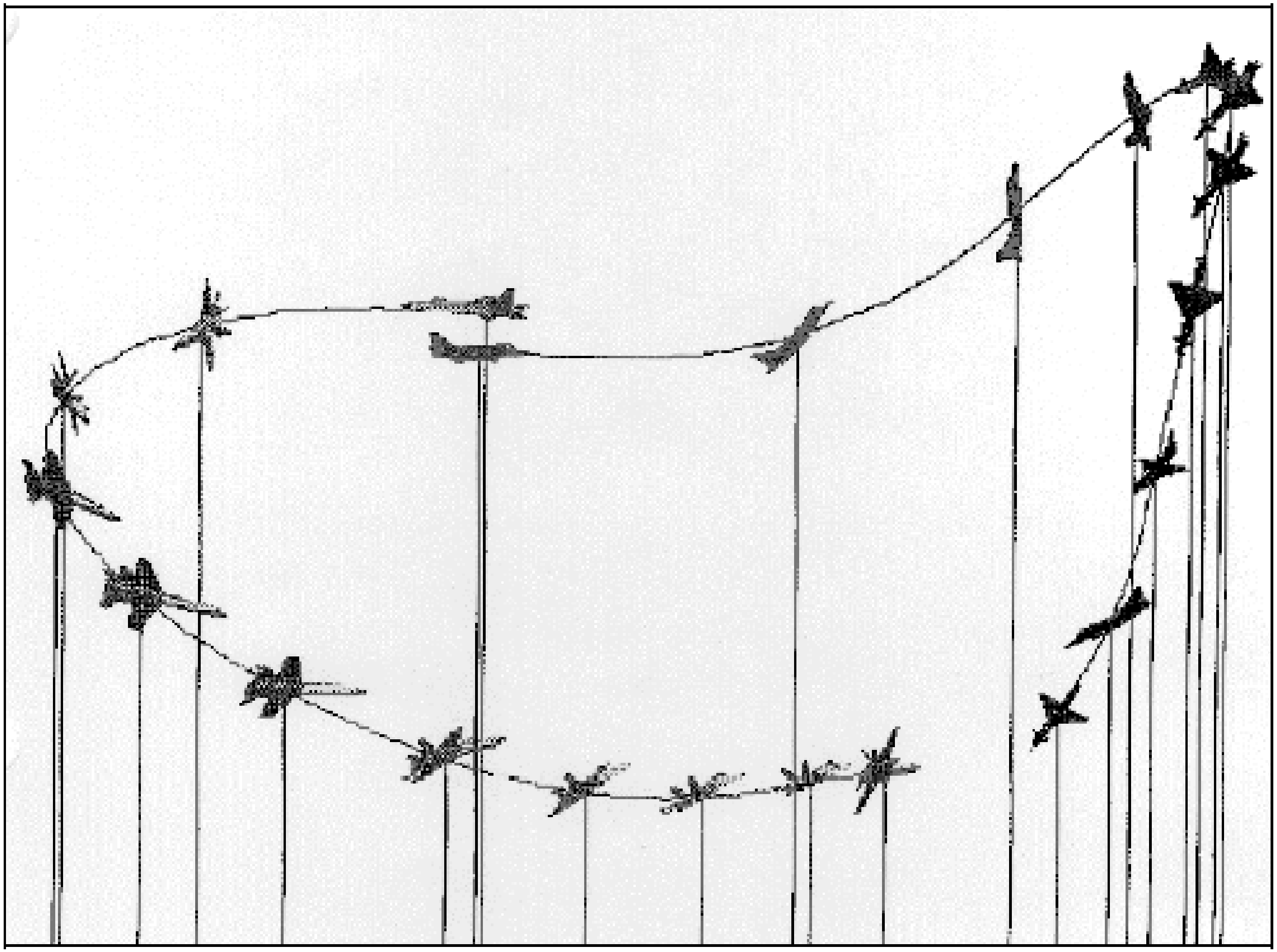
More like selective breeding than natural evolution

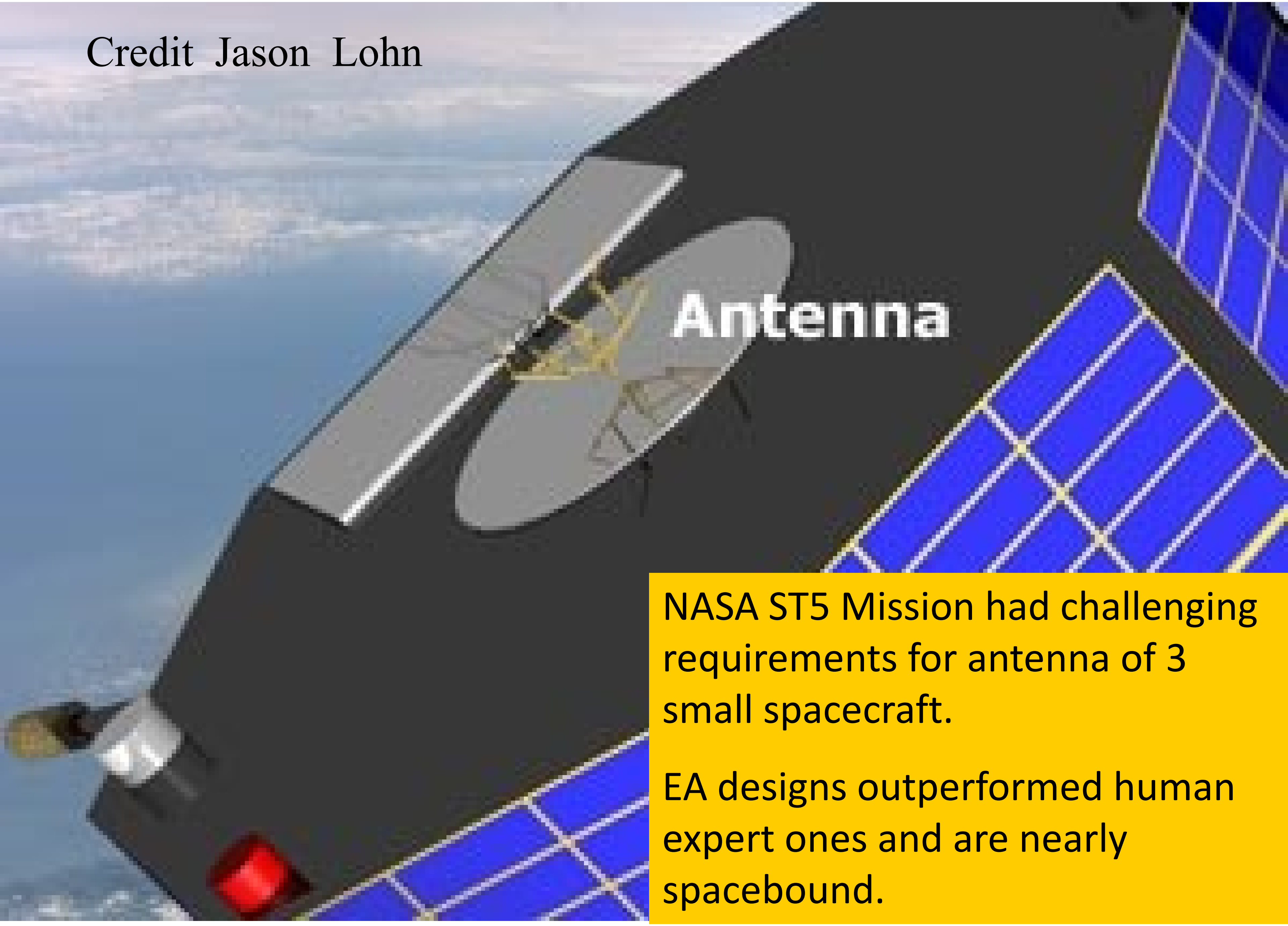


Evolving Top Gun strategies



Evolving Top Gun strategies

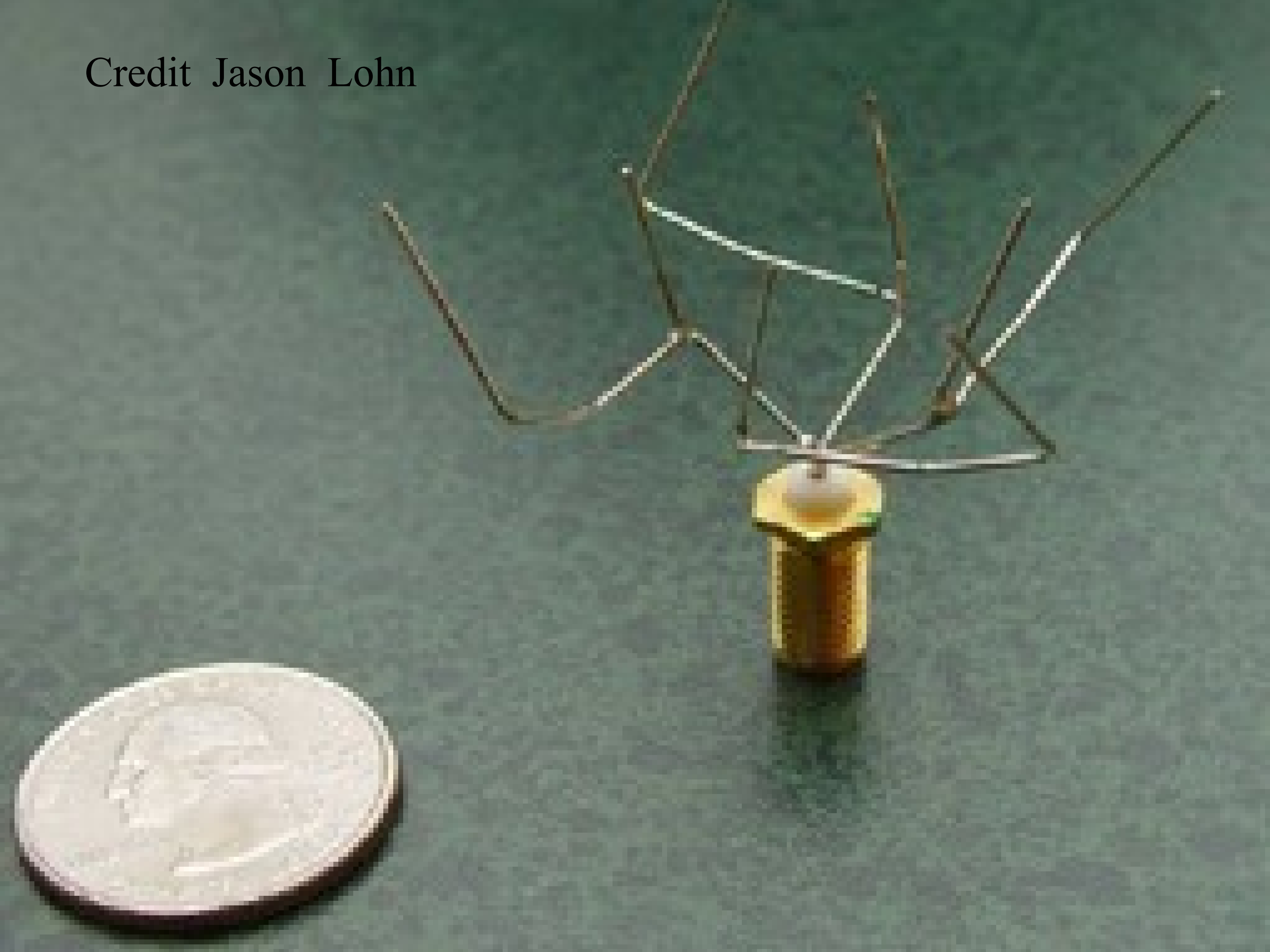




NASA ST5 Mission had challenging requirements for antenna of 3 small spacecraft.

EA designs outperformed human expert ones and are nearly spacebound.

Credit Jason Lohn



Application Areas

Planning

– Routing, Scheduling, Packing

Design

– Electronic Circuits, Neural Networks, Structure Design

Simulation

– Model economic interactions of competing firms in a market

Identification

– Fit a function to medical data to predict future values

Control

– Design a controller for gas turbine engine, design control system for mobile robots

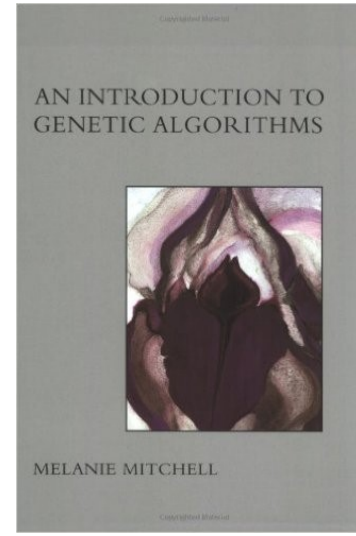
Classification

– Game playing, Diagnosis of heart disease, Detecting SPAM

Summary

- Module overview including assessment
- Introduced Nature Inspired Computation as a concept
- Basic principles & applications of evolutionary algorithms have been introduced

- Readings: An Introduction to Genetic Algorithms by M. Mitchell – Chapter 1



- Next Lecture: Varieties of Evolutionary Algorithms & Optimisation Problems